Intergenerational Economic Mobility and the Racial Wealth Gap

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Abstract

A growing body of research documents the importance of wealth and the racial wealth gap in perpetuating inequality across generations. We add to this literature by examining the impact of wealth on child income. Our two stage least squares regressions reveal that grandparental and parental wealth have an important effect on the younger generation's stock (first stage results), which in turn affects the younger generation's household income (second stage results). We further explore the relationship between income and wealth by decomposing the child's income by race. We find that the intergroup disparity in income is mainly attributable to differences in family background. These findings indicate that wealth is an important source of income inequality.

JEL Codes: D31, J15, J62, Z13

Keywords: intergenerational mobility, race, wealth

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1.0 Introduction

The conventional economic wisdom is that individual wealth accumulation is derived from savings out of current incomes. Such wisdom dates back to the *Wealth of Nations* by Smith (1776),[2] where Smith writes that "[p]arsimony [i.e. surplus of income over current consumption],…, is the immediate cause of the increase of [wealth]."[3] However, in this paper we raise the question of whether or not current incomes and economic well-being are linked to the previous generation's wealth. These links may be direct (inter-vivo transfers, inheritance), or indirect (social capital, peer effects from family social position).

Existing scholarship reveals that parental income is highly predictive of children's income (Corak 2013; Blanden et al. 2014). Wealth is, however, distinct from income.[4] Wealth, more than income, can be used to invest in appreciating assets for offspring, such as a college education, an unpaid internship in a high rent city, a new business, a property in a better residential neighborhood, or a job in the family firm (Corak and Piraino 2011). Without wealth transfers, regardless of income, these assets are harder to attain. Another strand of research explores the parent to adult children transmission of household wealth (Chiteji and Hamilton 2002; Charles and Hurst 2003); and the connections of three generations and the building blocks of net wealth, through the intergenerational transmission of risky, non-financial, and safe asset components (Toney 2020).

Less attention is focused on how the income of younger generations is influenced by the wealth of previous generations. According to Fox (2016), parental wealth propels a larger

[2] According to Schumpeter (1954, 186), Smith "marks the victory for more than 150 years of a pro-saving theory" by laying the conceptual foundation.
[3] In Chapter 3, "Of the Accumulation of Capital, or of the Productive and Unproductive Labour" (320).
[4] Net wealth is defined as assets minus liabilities, what is owned minus what is owed. Income is a flow of funds (paycheck from employment), while wealth is a stock of funds built from possession of assets (stocks, savings).

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proportion (62.1%) of white children above their parents' income position, compared to their black peers (42.4%). Parental homeownership, in particular, contributes to an even larger proportion (68.6%) of white children possessing income above their parents' income position, compared to their black counterparts (30.6%).[5]

The pursuit of wealth accumulation, through the propensity to save, may not be entirely confined to individuals and parents alone, but also may be dependent on grandparental wealth. Our 2SLS/IV regressions reveal that grandparental wealth has some effect on the third generation's stock (first stage results), which in turn affects the third generation's income (second stage results). We further explore the relationship between income and wealth by decomposing the child's income by race. Here, we find that the disparity in income between black and white respondents is mainly attributable to differences in family background. In context, differences in family background are stronger than differences in educational attainment. By including a three generation approach, we are able to take a longer term perspective on wealth, its accumulation, and its impact on economic mobility. We are also able to illustrate how stratification processes work across much longer periods of time, and highlight how inequality is perpetuated.

2.0 Data

The Panel Study of Income Dynamics (PSID) is appropriate for our research questions because we can explore extended family wealth and offspring economic status. The PSID survey data is representative of the U.S. population. Collected on all respondents, PSID content includes demographic data, e.g. gender, and family characteristics; and economic data, e.g. employment status, income, and wealth. Wealth information, in the form of assets and liabilities, was first

[5] Indeed, black Americans have lower upward mobility and higher downward mobility, compared to white Americans, and this is particularly true for black men because their families are not in a position to make intergenerational wealth transfers (Darity and Mullen 2020).

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collected in the 1984 wave. Moreover, the 2013 PSID's Roster and Transfer supplement allows us to explore economic transfers moving directly across family generations. Since we want to follow children over the life course of 1984 to 2017, we restrict the children's sample to individuals who were under the age of 21 and living with their parents in 1984. Our analytic sample is composed of 1,948 parent-child pairs, 1,032 paternal grandparent-grandchild pairs, and 1,107 maternal grandparent-grandchild pairs.

2.1 Summary Statistics

2.1.1 Income at Life Events

We find that a rise in median household income is associated with an advancement in life events, starting from forming their own household ($35,040), becoming a first-time homeowner ($60,000), completing a college education ($60,321), entering their first marital union ($64,774), having their firstborn child ($67,737), and peaking at turning forty years of age ($85,614). Meanwhile, aligned with life events is the black-white gap in median household income. When white children are forming their own households and becoming a first-time home buyer, black households possess sixty and seventy cents, respectively, for every dollar in income possessed by white households. When our black sample completes their college education and enters their first marriages, they hold about seventy and eighty cents, respectively, for every dollar amassed by our white sample. Finally, when white families and black families are having newborn children, black families earn sixty cents for every dollar earned by whites. When whites and blacks are turning forty years of age, blacks earn fifty cents for every dollar earned by whites.

2.1.2 Parental Transfers and Income Prospects

Receiving parental support for post-secondary educational attainment allows for a greater capacity to amass household income at the median ($100,380 versus $69,300 for those that did not receive support). Consistent with this trend on receiving parental support for post-secondary

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education is the pattern by race (for blacks, the median income is $66,150 versus $43,100 for those that did not receive support, while for whites the corresponding figures are $107,119 versus $86,310). Moreover, receiving parental support for homeownership is associated with higher incomes ($105,105 versus $73,505 for households that did not receive homeownership provisions). Evidence of this theme also is revealed in the black-white patterns ($84,032 versus $45,665 for the black sample; $110,250 versus $90,085 for whites).[6]

3.0 Empirical Strategy

Our empirical strategy measures the indirect effect of extended family wealth on income mobility. Equation (1) represents the first stage, while Equation (2) represents the second stage, of the two stage least squares (2SLS/IV) panel regression models,

(1) $$\mathrm{LnStocks}_{it} = C'_{it}\alpha + u_{it}$$

(2) $$\mathrm{LnInc}_{it} = X'_{it}\beta + \mathrm{LnStocks}'_{it}\phi + \varepsilon_{it}$$

where LnInc_{it} describes the inverse hyperbolic sine (log) transformation applied to household income for adult child i = 1,2,…,N, in survey year t = 1,2,…,T. X'_{it} is a vector of control variables for adult children, including age, age squared, and race (black=1). We instrument the log of stock value ($\mathrm{LnStocks}_{it}$) wth C'_{it} variables in the first stage.

Note, $\mathrm{LnStocks}_{it}$ is the inverse hyperbolic sine (log) transformation applied to stocks and mutual funds held outside of pension wealth for adult child i in survey year t. C'_{it} is a vector of control variables that include (log) parental wealth, (log) parental income, educational attainment, number of children in the household, and dichotomous variables for sex (male=1), marital status, employment status, professional or managerial occupation status, pension ownership,

[6] These income figures are not conditional on an adult child receiving post-secondary training or purchasing a home.

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homeownership, computer and internet use, risk avoidance, and time. Finally, u_{it} and ε_{it} are error terms associated with the first and second stage, respectively. A variant of Equation (1) and Equation (2) is run where (log) grandparental wealth and (log) grandparental income are variables of interest in the C'_{it} vector.

The rationale for such a 2SLS/IV model specification is based upon the mechanisms through which extended family wealth can affect the younger generation's income. Those mechanisms are both direct and indirect. Extended family wealth may be used to smooth income during offspring employment gaps or financial hardships (Fox 2016), which is a direct form of family insurance. With regards to investment, extended family wealth can help with the child's homeownership acquisition, higher educational attainment (Johnson 2020), or stocks and mutual funds investing (Chiteji and Stafford 1999; Toney 2020), which later on improve offspring income and wealth.

Models that allow us to isolate contributors to the intergroup differences in income include Oaxaca-Blinder decomposition techniques. The Oaxaca-Blinder decomposition is an instrument of isolation, through a reduced form analysis. Such a technique can provide a sense of how much of income mobility is anchored to extended family wealth, how much is driven by personal initiative, and how much is due to other social processes in the U.S. economy. We use the threefold Oaxaca-Blinder decomposition, which can isolate contributors into three components: endowment, coefficient, and interaction. The black-white income gap (log) is the outcome variable. What the endowment effect seeks to explain is the intergroup income gap that is attributable to control variables. An endowment effect can measure the percentage change in a black respondent's income had they possessed the equivalent family background and credentials as whites. The coefficient effect isolates the part of the income gap that is attributable to differences

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in slope parameters. Economists posit that this effect is analogous to discrimination. The interaction effect postulates that the endowment effect and differences in treatment can have a simultaneous effect on the income gap.

4.0 Empirical Results

Table 1 presents the second stage results from our 2SLS/IV panel regressions. Overall, our 2SLS/IV regressions reveal that parental and grandparental wealth have an important effect on the younger generation's stock (first stage results), which in turn affects the younger generation's income (second stage results). In the parent-child sample, the first stage estimates the effect of parental wealth on younger generation's stock, and the second stage shows that a 1% increase in stock value increases income by 0.25% (Column (1), Table 1). Within the paternal grandparent-grandchild sample, the first stage estimates the effect of grandparental wealth on younger generation's stock, and the second stage shows that a 1% increase in stock value increases income by 0.22 % (Column (2), Table 1). Finally, in the maternal grandparent-grandchild sample, the first stage estimates the effect of maternal grandparent wealth on younger generation's stock, and the second stage shows that a 1% increase in stock value increases income by 0.18% (Column (3), Table 1).

Given that race and household income are strongly negatively correlated (Table 1), we further explore this theme through the Oaxaca-Blinder decomposition technique (Table 3, parent-child pairs in Panel A, grandparent-grandchild pairs in Panel B).[7] In 2017, the black-white income gap (log) is 0.9206 in Panel A and is 0.6104 in Panel B (Column (1)). A theme of Table 3 is that a vast majority of the intergroup inequality in income is attributable to differences in endowments, including extended family wealth and income (Column (2)). In fact, long standing differences in

[7] Maternal grandparent lineage estimates are shown in Panel B of Table 3. However, we find fundamentally similar results when we compute the decomposition with the paternal grandparent lineage sample.

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family background account for two-tenths (0.2134 is attributable to parental income and wealth, and 0.1803 for grandparental income and wealth) of the black-white income gap. By contrast, differences in educational attainment account for a lower percentage, less than half of the impact of family background (0.1077 in Panel A; 0.1004 in Panel B). As the decomposition reveals in Column (3), the gaps in treatment translate into a substantial difference in income mobility (0.4243 in Panel A; 0.1068 in Panel B) for black respondents.[8]

5.0 Conclusion

In this paper, we have demonstrated that grandparental and parental wealth have some effect on the third generation's stock (first stage results), which in turn affects the third generation's adult income (second stage results). Such a finding indicates that wealth is an important predictor of later life income for both black and white Americans, and that the unequal distribution of wealth has long ranging consequences for the younger generation and inequality overall. Focusing on individual efforts and two generations will not provide a complete picture. Our analysis indicates not only that the prior generation's endowments are crucial to understanding the subsequent generations attainment, but also that the impact of those endowments are not constant across racial groups.

[8] As a robustness check, we examine different income percentiles. We find that the effect of grandparental and parental wealth endowment is much stronger at the top of the income distribution (90th percentile).

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Table 1: 2SLS/IV Coefficients of Stock Value Influenced by Parental and Grandparental Wealth on the Younger Generation's Income

Variables	(1)	(2)	(3)
Log stock value influenced by parental wealth	0.2523***		
	(0.0133)		
Log stock value influenced by paternal grandparent's wealth		0.2200***	
		(0.0191)	
Log stock value influenced by maternal grandparent's wealth			0.1826***
			(0.0130)
Age	0.0765***	0.0619***	0.1136***
	(0.0159)	(0.0214)	(0.0203)
Age Squared	-0.0009***	-0.0007**	-0.0013***
	(0.0002)	(0.0003)	(0.0003)
Race (black=1)	-0.1821***	-0.2036***	-0.2395***
	(0.0424)	(0.0643)	(0.0527)
Constant	9.8493***	10.1878***	9.3144***
	(0.2830)	(0.3973)	(0.3802)
Person-Year Observations	12,835	3,608	3,774
Number of Children	1,948	1,032	1,107

Notes: Each column reports the second stage results from two stage least squares (2SLS/IV) panel regressions. We instrument the log of stock values (stocks and mutual funds held outside of pension wealth) with the C variables in the first stage. Log of stock values are regressed on (grand)parental wealth, (grand)parental income, educational attainment, number of children in household, marital status, employment status, professional or managerial occupational status, pension ownership, home ownership, computer/internet use, risk avoidance, time dummies, sex (Table 2). The model identification is based on the rationale that parental and grandparent wealth can indirectly influence the younger generation's income. Inverse hyperbolic sine (log) transformation of the adult child's household income is the dependent variable in the second stage. Inverse hyperbolic sine (log) transformation of parental wealth (Column (1)), paternal grandparent's wealth (Column (2)), and maternal grandparent's wealth (Column (3)) is the instrument of interest. Cluster robust standard errors are in parenthesis. *** p<0.01, ** p<0.05, * p<0.1.

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Table 2: First Stage Results of Younger Generation's Stock Value Influenced by Parental and Grandparental Wealth

Variables	(1) Stock Value	(2) Stock Value	(3) Stock Value
Parent's Wealth	0.0293***		
	(0.0066)		
Paternal Grandparent's Wealth		0.0029	
		(0.0143)	
Maternal Grandparent's Wealth			0.0463***
			(0.0136)
Person-Year Observations	12,835	3,608	3,774
Number of Children	1,948	1,032	1,107
R-squared	0.1617	0.1881	0.1950

Notes: Each column reports the first stage results from two stage least squares (2SLS/IV) panel regressions, showing that grandparental and parental wealth have some effect on the third generation's stock value (stocks and mutual funds held outside of pension wealth). See Table 1 for details. *** p<0.01, ** p<0.05, * p<0.1.

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Table 3: Results from Oaxaca-Blinder Decomposition of Income Gap

Panel A: Parent-Child

	(1) Differential	(2) Endowments	(3) Coefficients	(4) Interaction
Total		0.5863***	0.4243***	-0.0900
		(0.0476)	(0.1219)	(0.1049)
White Household Income(log)	12.0467***			
	(0.0325)			
Black Household Income(log)	11.1261***			
	(0.0987)			
White-Black Income Gap	0.9206***			
	(0.1039)			
Observations	1,948	1,948	1,948	1,948

Panel B: Grandparent-Grandchild

	(1) Differential	(2) Endowments	(3) Coefficients	(4) Interaction
Total		0.4125***	0.1068	0.0911
		(0.0676)	(0.1153)	(0.0987)
White Household Income(log)	12.0003***			
	(0.0503)			
Black Household Income(log)	11.3898***			
	(0.0772)			
White-Black Income Gap	0.6104***			
	(0.0921)			
Observations	1,107	1,107	1,107	1,107

Notes: The reduced-form decomposition models of Panel A and Panel B contain the following covariates: educational attainment, number of children in household, age, age squared, marital status (married =1), sex (men=1), professional or manager occupational status (yes=1). Chief covariates of interest include extended family wealth (average (log) of parental wealth from 1984 and 1989 waves in Panel A; average (log) of grandparental wealth from 1984 and 1989 waves in Panel B) and extended family income (average (log) of parental income from 1984 and 1989 waves in Panel A; average (log) of grandparental income from 1984 and 1989 waves in Panel B). In Panel A, parental income and wealth variables may account for 0.2134 (0.1965/0.9206), and educational attainment may account for 0.1077 (0.0992/0.9206), of the black-white income gap. In Panel B, 0.1803 (0.1101/0.6104) of the black-white income gap may be attributed to grandparental income and wealth variables, while 0.1004 (0.0613/0.6104) may be attributed to educational attainment. The dependent variable is inverse hyperbolic sine (log) transformation applied to the child's household income. *** p<0.01, ** p<0.05, * p<0.1.

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